Analysis of the IPL Dataset

Mini Project 2 — Final Report (Part C)

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Data Science Summer Training – Batch 3

1. Executive Summary

The IPL has evolved into a significantly higher-scoring, more strategically chase-oriented tournament than it was at its inception. Across 17 seasons and over 260,000 deliveries, three forces consistently shape match outcomes: the toss decision (fielding first has been statistically favoured in most seasons since 2014), home-venue advantage (every top-5 stadium shows a dominant home franchise), and a clear long-term rise in scoring rates, culminating in 2024's record average of 365.8 runs per match. The single most actionable insight for a captain or coach is that **winning the toss and electing to field first has been the stronger strategic choice in the majority of recent seasons**, and team strategy should be built around chasing competently rather than defending a total.

2. Key Insights

Insight 1: IPL Is Becoming Significantly More High-Scoring

Average runs per match have risen from the 287–315 range in the early seasons (2008–2014) to a record 365.8 runs/match in 2024 — the highest in IPL history. A regression trend line confirms a clear, sustained upward slope rather than random fluctuation, rising from roughly 293 to 342 runs/match across the 17-season span. This is most plausibly driven by improving pitches, shorter boundaries, more aggressive 360-degree batting techniques, and rule changes such as the Impact Player rule introduced in 2023.

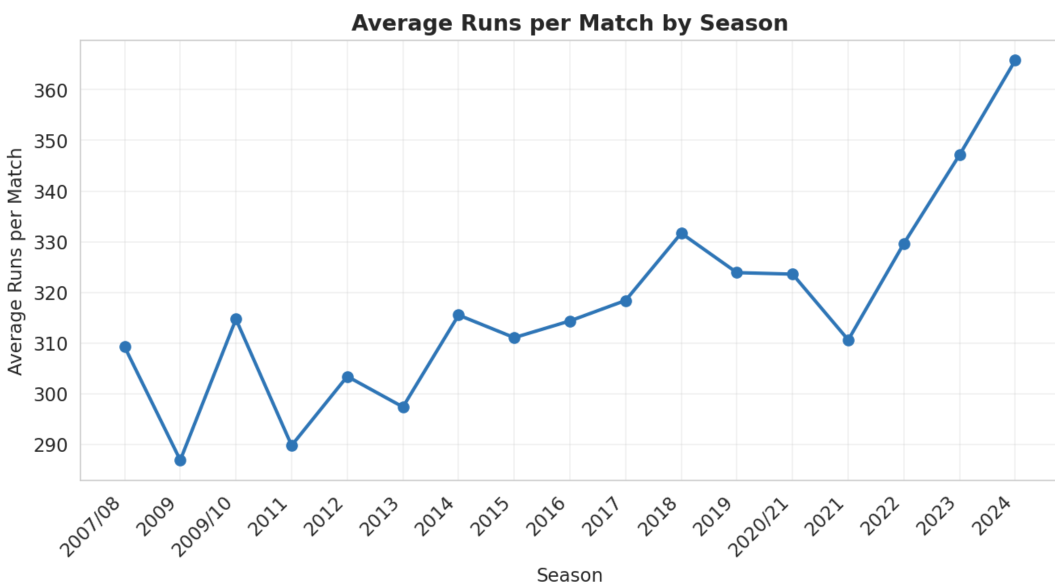

Figure 1: Average runs per match by season, 2007/08–2024.

Insight 2: Fielding First Has Become the Statistically Favoured Toss Decision

From 2014 onward, teams that won the toss and chose to field first won more often than those who chose to bat first in most seasons — for example, an 65% win rate for fielding in 2016 versus just 18% for batting that same season. A small number of seasons (2018, 2019, 2023) reversed this pattern, but the broader trend strongly favours chasing, consistent with the common cricketing narrative that dew affects bowling grip under lights and modern batting depth is well suited to chasing a known target.

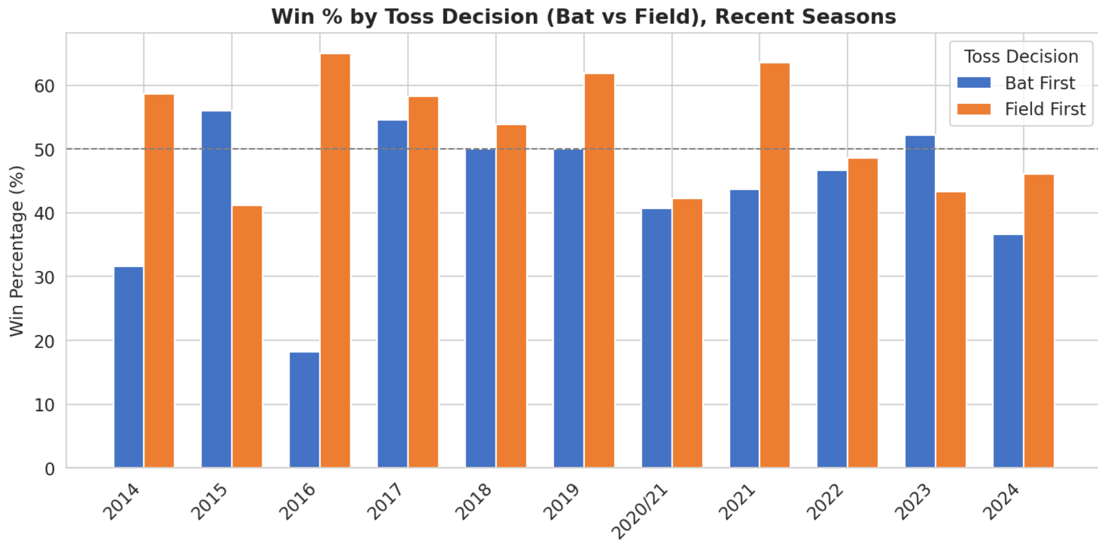

Figure 2: Win percentage by toss decision (bat vs. field), recent seasons.

Insight 3: A Small Group of Players Dominate All-Time Run-Scoring

Virat Kohli leads all-time IPL run-scoring with 8,014 runs, well clear of S Dhawan (6,769), RG Sharma (6,630), DA Warner (6,567), and SK Raina (5,536). The list is dominated by long-serving top-order batters and openers who face the highest ball volume, alongside durable middle-order anchors. Dismissal analysis of the top three (Kohli, Dhawan, Sharma) shows that all three are dismissed primarily by being caught (60–75% of dismissals), reflecting attacking intent rather than any shared technical weakness against a specific bowling style.

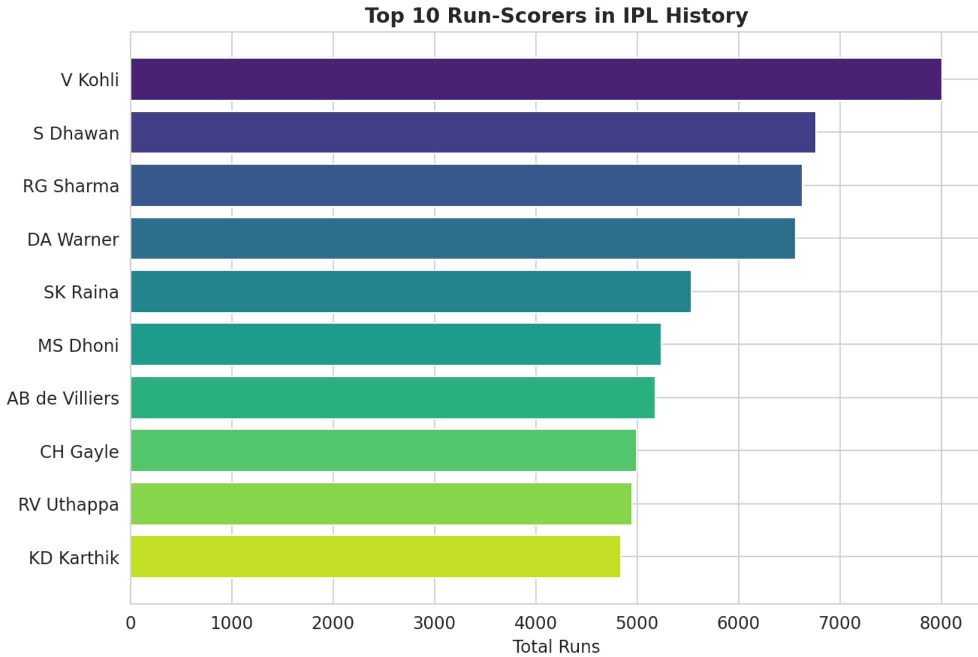

Figure 3: Top 10 all-time IPL run-scorers.

Insight 4: Home Venues Provide a Decisive Advantage

At every one of the top five most-used venues, the home franchise has the highest win count by a wide margin. Kolkata Knight Riders have won 45 matches at Eden Gardens — more than four times the next-best team's tally there — and Mumbai Indians have a similarly dominant 42 wins at Wankhede Stadium. This pattern, repeated across Chinnaswamy, the Rajiv Gandhi Stadium, and Feroz Shah Kotla, points to a consistent, measurable home-ground advantage likely rooted in pitch familiarity and crowd support.

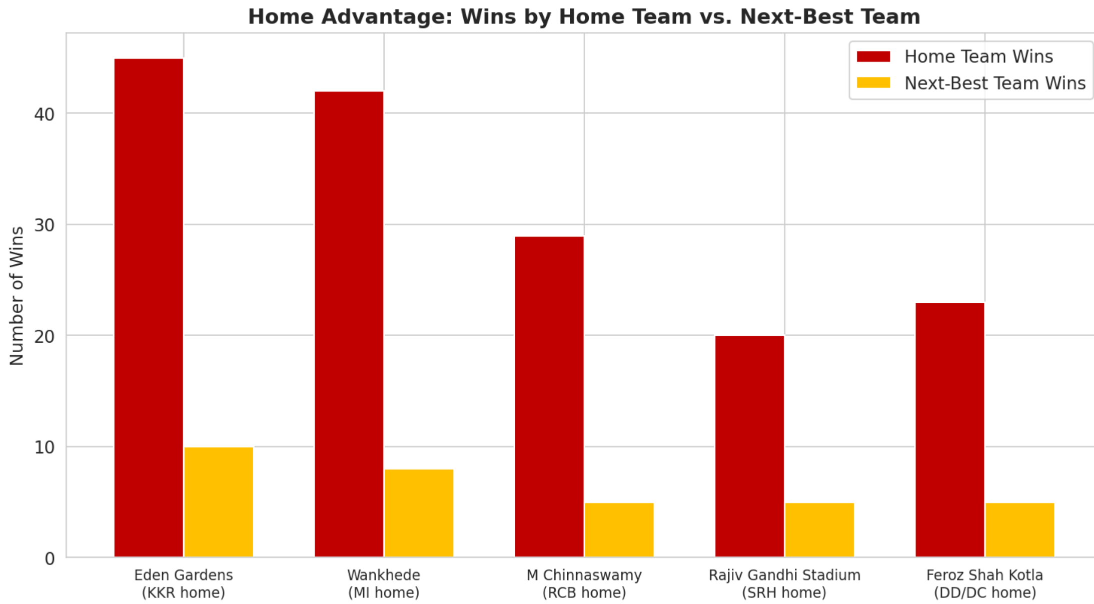

Figure 4: Home team wins vs. next-best team wins at the top 5 venues.

Insight 5: Chennai Super Kings Are the Most Explosive Death-Over Batting Side

Among the six highest death-over-scoring franchises, Chennai Super Kings post the highest median runs per over (10 vs. 9 for every other team) and the widest scoring range, including the single highest over total in the dataset (37 runs). Kings XI Punjab are the most consistent but least explosive, with the narrowest scoring range of the six. This suggests CSK's death-over approach is built around high-ceiling aggression, while teams like Kings XI Punjab favour lower-variance accumulation in the closing overs.

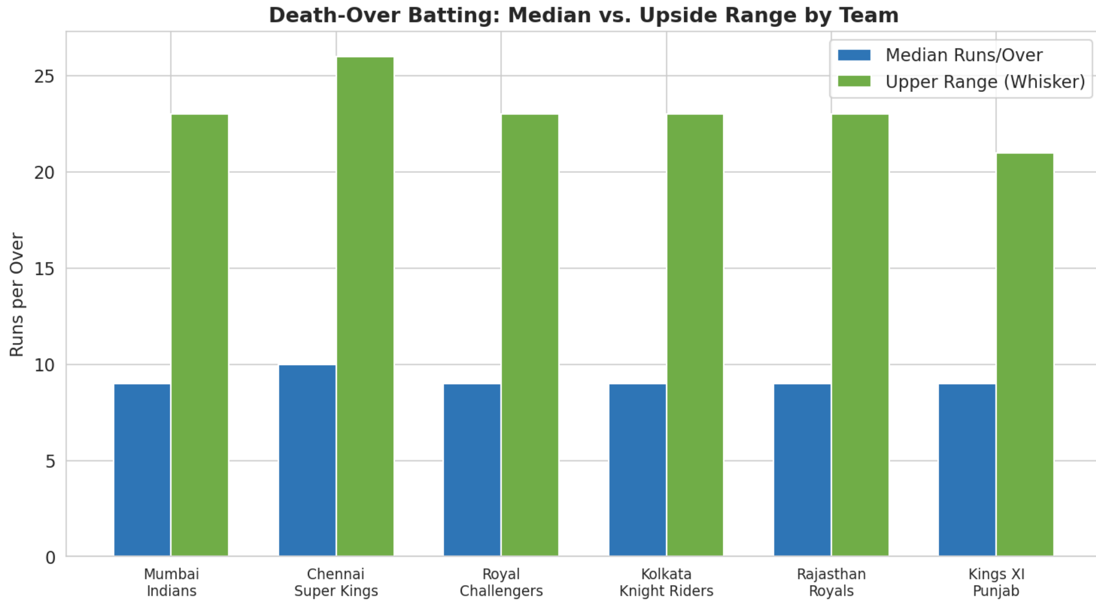

Figure 5: Median and upper-range runs per over in death overs (16–20), top 6 teams.

3. Recommendation for Team Strategy

Recommendation: prioritise winning the toss to field first, and build squad strategy around chasing, not defending. The data shows fielding-first has been the stronger toss decision in most seasons since 2014, with win-rate gaps as large as 47 percentage points in a single season (2016). A captain should treat a won toss as an opportunity to chase by default, deviating only when pitch reports, dew forecasts, or an unusually strong death-bowling attack of their own argue otherwise. Practically, this means recruiting and batting-order planning should favour players proven to perform under chase pressure with a clear target, rather than purely first-innings stroke-players. A secondary, supporting recommendation: teams should weight scheduling and squad-building decisions around their home venue's specific conditions, since the data shows home advantage is large and consistent across franchises.

4. Self-Reflection

A key limitation of this dataset is the absence of contextual variables that strongly influence match outcomes but are not captured here — specifically pitch reports, weather and dew conditions, and player fitness/injury status at the time of each match. The toss-and-venue analysis in this report identifies strong correlations (e.g., fielding-first winning more often, certain venues favouring certain teams), but without pitch and weather data, it is difficult to fully separate the toss decision's causal effect from confounding factors like dew or ground-specific dimensions. Additional data on player workload, recent form (e.g., runs/wickets in the preceding 5 matches), and head-to-head matchup history between specific batters and bowlers would allow for a more granular, predictive analysis rather than the descriptive, season-level patterns presented here.

End of Report — Harnoor Kaur, Mini Project 2